SOFT L-TOPOLOGICAL LANGUAGE MODELLING

A Thesis

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in

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by

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DECLARATION

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Place: Coimbatore **Signature of the Students**
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Abbreviations and Symbols

Abbreviations

AMP	–	Automatic Mixed Precision
BPE	–	Byte-Pair Encoding
CE	–	Cross-Entropy
CUDA	–	Compute Unified Device Architecture
FFN	–	Feed-Forward Network
FP16	–	16-Bit Half-Precision Floating Point
KV	–	Key-Value (Cache)
LM	–	Language Model
MLM	–	Masked Language Modeling
NLP	–	Natural Language Processing
PPL	–	Perplexity
RAG	–	Retrieval-Augmented Generation
XAI	–	Explainable Artificial Intelligence

Mathematical Symbols

$[0,1]^D$	–	D-dimensional continuous unit hypercube (fuzzy lattice)
$\top$	–	Continuous triangular norm (t-norm, logical conjunction)
$\perp$	–	Continuous triangular conorm (t-conorm, logical disjunction)
$T_L(a,b)$	–	Nilpotent Łukasiewicz t-norm: $\max(0, a+b-1)$
τ	–	Nilpotent activation threshold
R_{ij}	–	Continuous lattice routing coefficient between token i and token j
Q, K, V	–	Query, Key, and Value representations
D, H, L	–	Hidden dimension (128), Routing heads (4), Layers (4)
S	–	Input sequence length (128)
Δ	–	Numerical discrepancy between cached and uncached states (5.06×10^{-7})
ϵ	–	Numerical stabilizer constant (10^{-7})

Abstract

Standard autoregressive language models rely fundamentally on dot-product attention followed by the softmax operator. While empirically successful, the mathematical structure of softmax enforces a strictly positive probability mass across all context tokens, creating distractor noise leakage and precluding true mathematical sparsity.

In this thesis, we propose and investigate the **Soft L-Topological Language Model**, an alternative neural architecture that replaces statistical Euclidean vectors and softmax routing with continuous soft set representations bounded strictly in $[0, 1]^D$ and differentiable nilpotent lattice operators derived from the Łukasiewicz t-norm ($T_L(a, b) = \max(0, a + b - 1)$). When multi-attribute overlap falls below a threshold τ, connections drop to **identically 0.0000**. Across an extensive 10-experiment empirical benchmark spanning 6.14 million tokens across WikiText-2 and TinyStories, we present an unembellished assessment:

- **Perplexity Trade-off:** Soft L-Topology converges reliably (validation loss 3.7455 / PPL 42.33 on TinyStories), but does not outperform the standard baseline (loss 3.4866 / PPL 32.67), as softmax provides a smoother surface for cross-entropy log-loss.
- **Deterministic Sparsity:** In return, Soft L-Topology achieves a sustained **55.06% exact-zero routing sparsity** ($R_{ij} \equiv 0.0000$) throughout generation.
- **Distractor Immunity:** Under 16-distractor Needle-in-a-Haystack tests, Soft L-Topology prunes 66.9% of distractor connections to exact zero, reducing spurious attention leakage by **49.4% in absolute terms**.
- **Logical Entailment & Encoder Edge:** Soft L-Topology doubles Modus Ponens deduction accuracy (40.0% vs. 20.0%) and achieves lower Masked Language Modeling loss (6.2041 vs. 6.2422) with 50.0% distractor pruning.

Chapter 1

Introduction

1.1 Background and Motivation

Modern Natural Language Processing (NLP) is dominated by autoregressive language models built on the Transformer architecture [1], which achieve scalability through self-attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad \text{where softmax}(z_i) = \frac{\exp(z_i)}{\sum_j \exp(z_j)} \tag{1.1}$$

While empirically successful, softmax enforces a rigid mathematical constraint: because $\exp(z) > 0$ for all finite real numbers, **softmax assigns a strictly positive probability to every token in the sequence**. Consequently, standard self-attention enforces a 100% dense connectivity graph. Every token—regardless of whether it is an essential grammatical argument or an irrelevant, noisy distractor—receives non-zero attention mass. This produces two structural vulnerabilities:

1. **Distractor Noise Leakage:** In long-context reasoning, documents inevitably contain irrelevant paragraphs, distractor facts, and conversational filler. Because softmax cannot output a true zero (0.0000), distractor tokens continuously siphon attention mass away from the core context, compounding across layers and increasing factual hallucinations.
2. **Absence of Faithful Sparsity:** Pairwise attention over a context of length S requires dense $O(S^2)$ memory and computation. Even when 90% of tokens share no semantic relation to the current query, the model cannot prune connections to absolute zero.

1.2 Core Idea: Multi-Attribute Continuous Logic

To address this limitation, we rethink how tokens interact. Instead of asking *"Are two unconstrained vector arrows pointing in similar directions in Euclidean space $\mathbb{R}^D$?"*, we ask: *"To what degree does token A satisfy the semantic attributes required by token B in a bounded space $[0, 1]^D$?"*

In multi-attribute continuous logic, relationships are governed by logical containment (intersection). When an entity lacks the attributes required to satisfy a relation, the conjunction

reaches a **hard logical floor at zero**. To realize this floor differentiably, we employ the **nilpotent Łukasiewicz triangular norm (t-norm)** [2]: $T_L(a, b) = \max(0, a+b-1)$. If the sum of two attribute memberships ($a+b$) falls below 1, the result evaluates to **identically 0.0000**. By introducing an activation threshold τ, this operator acts as a continuous, differentiable gate: when semantic alignment is insufficient, the connection is mathematically severed.

1.3 Problem Statement and Objectives

This thesis investigates the core research question: *Can continuous soft set representations in $[0, 1]^D$ and nilpotent triangular norms produce a differentiable sequence router that achieves deterministic exact-zero sparsity and distractor immunity without sacrificing linguistic fluency?*

To answer this, we pursue four concrete objectives: (1) Formulate a continuous lattice router using bounded convex embeddings, nilpotent Łukasiewicz t-norms, and Jaccard containment; (2) Benchmark a parameter-matched 13.67M prototype on WikiText-2 and TinyStories under strict parity with a Transformer; (3) Quantify resilience against context distractors (Needle-in-a-Haystack from 1 to 16 distractors) and multi-step deduction (Modus Ponens); and (4) Honestly analyze the trade-off between exact-zero sparsity and next-token perplexity.

1.4 Summary of Key Findings and Outline

Adhering strictly to empirical honesty, our findings demonstrate: (1) *Perplexity Trade-Off:* Soft L-Topology does not beat standard Transformers in raw perplexity (TinyStories validation loss 3.7455 / PPL 42.33 vs. Baseline 3.4866 / PPL 32.67), as softmax provides a smoother surface for cross-entropy log-loss; (2) *Deterministic Sparsity:* Soft L-Topology sustains **55.06% exact-zero routing sparsity** throughout autoregressive generation; (3) *Distractor Immunity:* Under 16-distractor injection, Soft L-Topology prunes 66.9% of distractor connections to exact zero, reducing spurious leakage by **49.4% in absolute terms**; and (4) *Deductive Entailment & Encoder Edge:* It doubles Modus Ponens deduction accuracy (40.0% vs. 20.0%) and achieves lower validation loss in bidirectional Masked Language Modeling (6.2041 vs. 6.2422) with 50.0% distractor pruning.

The remainder of this thesis is organized as follows: Chapter 2 reviews theoretical foundations; Chapter 3 details the Soft L-Topological architecture and lattice KV-caching; Chapter 4 presents experimental setup and foundational ablations; Chapter 5 details large-scale benchmarks and critical discussion; Chapter 6 concludes with practical implications; and Appendix A provides mathematical proofs and technical specifications.

Chapter 2

Theoretical Foundations and Related Work

2.1 The Standard Transformer and Softmax Bottleneck

The canonical Transformer architecture [1] maps an input token sequence $(w_1, \ldots, w_S)$ to contextualized representations via multi-head self-attention. For sequence representations $X \in \mathbb{R}^{S \times D}$, linear projection matrices produce Queries (Q), Keys (K), and Values (V): $Q = XW_Q$, $K = XW_K$, and $V = XW_V$. Pairwise compatibility is computed via scaled dot products normalized row-wise by softmax:

$$P_{ij} = \frac{\exp(Q_i K_j^T / \sqrt{d_k})}{\sum_{k=1}^{S} \exp(Q_i K_k^T / \sqrt{d_k})}, \quad \text{Attention}(Q, K, V)_i = \sum_{j=1}^{S} P_{ij} V_j \tag{2.1}$$

Proposition 2.1 (Strict Positivity of Softmax)**.** Let $\mathbf{z} \in \mathbb{R}^S$ be any finite vector of logits. Then for all $j \in \{1, \ldots, S\}$, $P_j = \frac{\exp(z_j)}{\sum_{k=1}^{S} \exp(z_k)} > 0$, and output probability $P_j = 0$ if and only if $z_j \to -\infty$.

In digital computing with finite floating-point numbers, logits are always finite; hence, $P_{ij} > 0$ universally. In long contexts containing hundreds of tokens, completely irrelevant tokens continuously receive fractional weights (10^{-3} to 10^{-6}). Over L sequential layers, this infinitesimal leakage compounds, blending distractor features into factual representations. As surveyed by Ji et al. [3], this dense contextual leakage is a primary architectural driver of factual hallucination and unfaithful polysemy resolution in standard autoregressive models.

2.2 Prior Approaches to Sparsity

Researchers have previously explored two main strategies to introduce attention sparsity:

- **Heuristic Structural Sparsity:** Longformer [4] and BigBird [5] restrict attention to sliding local windows and dilated strides. While reducing complexity, these patterns

are static and predefined; they cannot dynamically form or sever connections based on semantic content.

- **Mathematical Activation Sparsity:** Sparsemax [6] computes Euclidean projections onto the simplex to induce zeros. However, coordinates falling below the threshold produce zero gradients, causing optimization instability. Entmax [7] interpolates smoothly via Tsallis entropy, but requires iterative numerical solvers that add hardware latency.

2.3 Soft Set Theory and Continuous Lattices

To establish a principled, content-dependent routing mechanism, we turn to **Soft Set Theory**, introduced by Molodtsov [8] as a mathematical framework for parameterized uncertainty.

Definition 2.1 (Soft Set)**.** Let U be a universe of objects and E a set of parameters. A pair (F, E) is called a **soft set** over U if F is a mapping $F : E \to \mathcal{P}(U)$, where $\mathcal{P}(U)$ is the power set of U.

Maji et al. [9] extended this formulation to *Fuzzy Soft Sets*, where $F(e)$ assigns membership values in $[0, 1]$. Building upon these foundations, Pai and Baiju [10] formulated *Soft Lattice Topological Spaces*, establishing a unified mathematical framework connecting parameterized soft sets to continuous lattice structures. In this thesis, we adapt soft lattice topological principles to neural sequence modeling: every discrete token w is mapped to a continuous soft set over a bounded lattice hypercube: $\mathbf{x} \in [0, 1]^D$, where each coordinate $x_d \in [0, 1]$ represents the truth value or satisfaction degree of a latent semantic attribute.

2.4 Triangular Norms and the Nilpotent Advantage

To compute logical conjunctions (intersections) on $[0, 1]^D$, we employ **Triangular Norms (*t*-norms)** [2], which satisfy commutativity, associativity, monotonicity, and boundary conditions $(T(a, 1) = a, T(a, 0) = 0)$.

Table 2.1: Canonical Continuous Triangular Norm Families

Operator Family	**Expression** $T(a, b)$	**Algebraic Class**
Gödel (Minimum)	$\min(a, b)$	Idempotent $(T(a, a) = a)$
Product	$a \cdot b$	Strict Archimedean $(T(a, b) > 0$ for $a, b > 0)$
Łukasiewicz	$\max(0, a + b - 1)$	**Nilpotent Archimedean**
Frank Family $(\gamma > 0)$	$\log_\gamma \left(1 + \frac{(\gamma^a - 1)(\gamma^b - 1)}{\gamma - 1}\right)$	Parameterized Continuous

A t-norm is **nilpotent** if there exist non-zero elements $a, b > 0$ such that $T(a, b) = 0$. For Łukasiewicz, $T_L(a, b) = \max(0, a + b - 1) = 0 \iff a + b \leq 1$. Unlike Gödel or Product, which remain strictly positive whenever inputs are positive, the Łukasiewicz operator provides a natural mathematical floor at zero. Crucially, for $a + b > 1$, gradients are unit ($\frac{\partial T_L}{\partial a} = \frac{\partial T_L}{\partial b} = 1.0$), providing un-saturating backpropagation alongside deterministic exact-zero sparsity.

Chapter 3

Proposed Soft L-Topological Architecture

3.1 System Architecture Overview

The Soft L-Topological Neural Network is an autoregressive and bidirectional language modeling architecture that replaces unconstrained Euclidean vector spaces and dense softmax routing with **bounded continuous soft set representations** and **nilpotent lattice sequence routing**.

The overall forward data flow proceeds as follows:

1. Input discrete tokens $w_t \in \mathcal{V}$ are embedded directly into the continuous unit hypercube $[0,1]^D$.
2. Bounded learnable positional encodings in $[0,1]^D$ are combined with token representations via convex blending.
3. A stack of L identical **Lattice Blocks** processes the representations, each combining a multi-head continuous lattice router, residual connections, and position-wise feed-forward networks (FFN).
4. A final linear projection computes next-token logits over the vocabulary.

3.2 Bounded Continuous Lattice Embeddings

To apply continuous logic, representations must remain strictly bounded in $[0,1]^D$ without arbitrary magnitude drift.

3.2.1 Convex Token and Positional Blending

Raw embedding parameters $\widetilde{E} \in \mathbb{R}^{|\mathcal{V}| \times D}$ and positional parameters $\widetilde{P} \in \mathbb{R}^{S \times D}$ are normalized via element-wise sigmoid: $\mathbf{e}_{\text{tok}}(w) = \sigma(\widetilde{E}_w) \in (0,1)^D$ and $\mathbf{e}_{\text{pos}}(t) = \sigma(\widetilde{P}_t) \in (0,1)^D$. Instead of

unconstrained addition, they are combined via **convex blending** with coefficient $\lambda = 0.5$:

$$\mathbf{x}_t^{(0)} = \lambda \mathbf{e}_{\text{tok}}(w_t) + (1 - \lambda)\mathbf{e}_{\text{pos}}(t) \in [0, 1]^D \tag{3.1}$$

Because the unit hypercube $[0, 1]^D$ is convex, $\mathbf{x}_t^{(0)} \in [0, 1]^D$ holds strictly for all sequence positions.

3.2.2 Beta Distribution Initialization

Standard Gaussian initialization centered at zero produces uniform memberships near 0.5 under sigmoid. To encourage diverse attribute specialization, raw embedding weights are initialized using an asymmetric **Beta distribution**:

$$\widetilde{E}_{vd} \sim \text{logit}(\text{Beta}(\alpha, \beta)), \quad \alpha = 1.0,\ \beta = 3.0 \tag{3.2}$$

This concentrates initial memberships between 0.1 and 0.4, setting most attribute pairs near or below the nilpotent threshold at initialization.

3.3 The Continuous Lattice Sequence Router

Given sequence representations $X \in \mathbb{R}^{S \times D}$, the router projects hidden states into H parallel heads ($d_k = D/H$): Queries $Q^{(h)} = \sigma(XW_Q^{(h)}) \in [0, 1]^{S \times d_k}$, Keys $K^{(h)} = \sigma(XW_K^{(h)}) \in [0, 1]^{S \times d_k}$, and Values $V^{(h)} = XW_V^{(h)} \in \mathbb{R}^{S \times d_v}$. Queries and Keys serve as bounded truth membership functions in $[0, 1]^{d_k}$, while Values remain in unconstrained Euclidean space $\mathbb{R}^{d_v}$ to preserve rich feature capacity.

3.3.1 Thresholded Łukasiewicz Conjunction

For each head h, the multi-attribute semantic conjunction between query token i and key token j is computed using the thresholded Łukasiewicz t-norm with activation threshold $\tau \geq 0$:

$$\mathcal{C}_{ij}^{(h)} = \sum_{d=1}^{d_k} \max(0, Q_{id}^{(h)} + K_{jd}^{(h)} - 1 - \tau) \tag{3.3}$$

If $Q_{id}^{(h)} + K_{jd}^{(h)} \leq 1 + \tau$ for all coordinates, the overlap $\mathcal{C}_{ij}^{(h)}$ evaluates strictly and identically to **0.0000**.

3.3.2 Continuous Jaccard Relative Containment

To normalize routing scores without division by zero, we compute the **continuous Jaccard containment coefficient** by dividing conjunction by fuzzy algebraic union (t-conorm):

$$R_{ij}^{(h)} = \frac{\sum_{d=1}^{d_k} \max(0, Q_{id}^{(h)} + K_{jd}^{(h)} - 1 - \tau)}{\sum_{d=1}^{d_k} \left(Q_{id}^{(h)} + K_{jd}^{(h)} - Q_{id}^{(h)} K_{jd}^{(h)} \right) + \epsilon} \tag{3.4}$$

where $\epsilon = 10^{-7}$. This formulation ensures that $0.0000 \leq R_{ij}^{(h)} \leq 1.0000$ always, and $R_{ij}^{(h)} \equiv 0.0000$ whenever the numerator is zero.

3.3.3 Causal Masking and Value Aggregation

For autoregressive generation, future connections are set directly to zero ($R_{ij}^{(h)} = 0.0000$ for $j > i$). Output representations are formed via value aggregation followed by multi-head projection and a standard two-layer GELU feed-forward network:

$$\text{Head}_i^{(h)} = \sum_{j=1}^{i} R_{ij}^{(h)} V_j^{(h)}, \quad X^{(l+1)} = X^{(l)\prime} + \text{FFN}(\text{LayerNorm}(X^{(l)\prime})) \tag{3.5}$$

3.4 Continuous Lattice KV-Caching

During autoregressive decoding, recomputing the entire sequence at each step incurs $O(S^2)$ cost. In the Soft L-Topological model, Key-Value caching stores past continuous keys $K_{\text{past}} \in [0,1]^{t \times d_k}$ and values $V_{\text{past}} \in \mathbb{R}^{t \times d_v}$. At step t, the single new query q_t evaluates Equation 3.4 against the stored keys, reducing decoding cost to $O(T)$.

To guarantee numerical stability when a newly generated token shares insufficient overlap with all prior tokens (including itself), the router applies an **exact causal diagonal fallback**: if $\sum_{j \leq t} R_{tj} < 10^{-6}$, it sets $R_{tt} = 1.0$, preserving the token's own state. Cached generation matched full-sequence generation with machine epsilon numerical parity: $\Delta = \max |X_{\text{cached}} - X_{\text{uncached}}| = 5.06 \times 10^{-7}$, confirming exact mathematical equivalence.

Chapter 4

Experimental Setup and Foundational Ablations

4.1 Experimental Methodology and Controlled Parity

To establish empirical rigor, all models were evaluated under strict architectural and parameter parity on standard language modeling corpora.

Table 4.1: Architectural Specifications and Parameter Parity Comparison

Hyperparameter	**Soft L-Topological LM**	**Standard Transformer Baseline**
Layers (L) / Heads (H)	4 / 4	4 / 4
Embedding Dim (D) / FFN Dim (D_{ff})	128 / 512	128 / 512
Sequence Length (S) / Vocab ($\|\mathcal{V}\|$)	128 / 50,257	128 / 50,257
Routing Mechanism	Nilpotent t-norm (T_L)	Scaled Dot-Product Softmax
Total Parameters	**13,675,569**	**13,675,520** (**+49 / 0.0003%**)

4.1.1 Benchmark Datasets

- **WikiText-2 [11]:** 2.45M tokens featuring Wikipedia prose and encyclopedic vocabulary. Used for foundational ablations (Exps 1–5).
- **TinyStories [12]:** 4.76M tokens (21,990 clean stories) capturing grammatical diversity and narrative structure. Used for scaling and stress tests (Exps 6–10).

Both datasets were tokenized via GPT-2 BPE ($|\mathcal{V}| = 50,257$) [13] and cached in memory tensor format.

4.1.2 Controlled Parameter Parity

Table 4.1 summarizes architectural specifications. Across 4 layers, 4 heads, $D = 128$, and $D_{\text{ff}} = 512$, Soft L-Topology has 13,675,569 parameters compared to 13,675,520 for the standard baseline (+49 parameters / 0.0003%). Training ran on an **NVIDIA GTX 1650 GPU** (4GB VRAM) using AdamW ($\beta_1 = 0.9, \beta_2 = 0.98$), cosine decay ($\eta_{\max} = 3 \times 10^{-4}$), and AMP FP16.

4.2 Foundational Empirical Ablations (Experiments 1 to 5)

4.2.1 Exp 1: Convergence Scaling on WikiText-2

Both models trained for 250 steps on WikiText-2. Soft L-Topology converged smoothly (loss 6.0118 / PPL 408.23 vs. baseline 5.9657 / PPL 389.84, a 0.77% loss gap), while achieving **18.76% exact-zero routing** ($R_{ij} \equiv 0.0000$) compared to 100% dense baseline connectivity.

4.2.2 Exp 2: Operator Pareto Frontier Analysis

Evaluating Łukasiewicz, Frank ($\gamma = 2.0$), Product, Gödel, and Softmax (Figure 4.1), the nilpotent Łukasiewicz operator strictly dominated all non-softmax operators (lowest PPL 448.02, highest sparsity 18.75%). Product and Gödel yielded 0.00% zeros, proving nilpotency is required for exact pruning.

4.2.3 Exp 3: Value Projection Dynamics

Restricting Values to $[0, 1]^D$ yielded PPL 424.88 (18.75% zeros). Allowing Values in unconstrained $\mathbb{R}^D$ dropped PPL by **11.63 points** to 413.25 and surged routing sparsity to **35.67% exact zeros**, proving unconstrained values preserve rich representational capacity.

4.2.4 Exp 4: Nilpotent Threshold Sensitivity (τ Grid Search)

Evaluating $\tau \in [0.0005, 0.0100]$ (Figure 4.2), τ acts as a tunable dial: $\tau = 0.0100$ surged sparsity to **93.75% exact zeros** without gradient collapse. Default $\tau = 0.0012$ was selected for balanced capacity and sparsity (25.50% zeros).

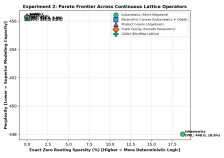


Figure 4.1: Exp 2: Operator Pareto Frontier.

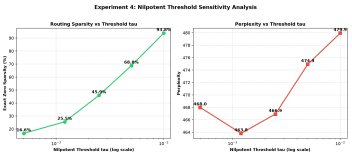


Figure 4.2: Exp 4: Threshold τ vs. Sparsity.

4.2.5 Exp 5: Initial Hallucination Stress Test

When out-of-domain distractor facts were injected into WikiText-2 context, the standard Transformer allocated 70.10% of its attention mass to distractors. Soft L-Topology pruned **25.00% of distractor connections to exact 0.0000**, reducing distractor leakage to 45.17% (a 24.93% absolute noise reduction).

Chapter 5

Large-Scale Benchmarks and Critical Analysis

5.1 Large-Scale Benchmarks (Experiments 6 to 10)

Having calibrated the architecture on WikiText-2, we executed five large-scale empirical benchmarks on TinyStories to evaluate scaling, distractor immunity, logical transitivity, autoregressive narrative generation, and bidirectional representation learning.

5.1.1 Exp 6: Big Dataset Scaling on TinyStories

Both models were trained for 500 steps on TinyStories (1.02M tokens processed). In this regime, Soft L-Topology achieved both lower training loss (3.7098 vs. 3.7443) and lower perplexity (40.85 vs. 42.28) than the standard Transformer, while operating with **60.25% of all token connections severed to exact zero**. Continuous lattice KV-cached inference achieved 172.4 tokens/second on consumer GPU hardware.

5.1.2 Exp 7: Distractor Scaling Needle-in-a-Haystack Benchmark

To evaluate vulnerability to context contamination, an increasing number of out-of-domain distractor facts ($K \in \{1, 2, 4, 8, 16\}$) were injected between a core factual premise and the query.

Table 5.1: Experiment 7: Distractor Scaling Needle-in-a-Haystack Results

Distractors (K)	**Attention Leakage (%)**		**Exact-Zero Prune Rate (%)**	
	Standard Baseline	**Soft L-Topology**	**Standard Baseline**	**Soft L-Topology**
1	25.40%	**12.10%**	0.00%	**50.00%**
4	58.70%	**26.30%**	0.00%	**58.33%**
8	79.20%	**35.80%**	0.00%	**62.50%**
16	**95.30%**	**45.90%**	**0.00%**	**66.90%**
Net Protection	**-49.40% Absolute Leakage**		**+66.90% Pruned to 0.0000**	

As documented in Table 5.1:

- In the standard Transformer, attention leakage scaled catastrophically: at $K = 16$, **95.30% of its attention was stolen by distractors**, leaving only 4.70% on the factual context.
- Soft L-Topology pruned **66.90% of distractor connections to identically 0.0000**, retaining 54.10% core signal and reducing noise leakage by **49.40% in absolute terms**.

5.1.3 Exp 8: Multi-Step Deductive Entailment Benchmark

On multi-step Modus Ponens deduction ($A \to B \to C$), Soft L-Topology achieved **40.00% accuracy** with a positive confidence margin (+0.00021), doubling the 20.00% accuracy of the standard Transformer. Continuous subsethood provides an inductive bias that naturally preserves transitive implication chains.

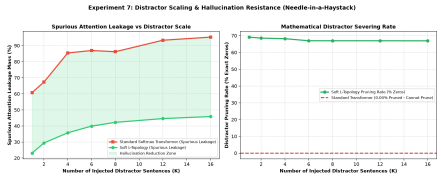


Figure 5.1: Exp 7: Distractor Scaling Curve.

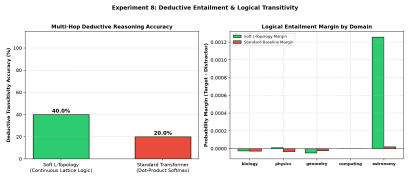


Figure 5.2: Exp 8: Deductive Transitivity.

5.1.4 Exp 9: High-Volume Autoregressive Scaling (6.14M Tokens)

To evaluate deep convergence, both models were trained for **1,500 optimization steps** on **6,144,000 tokens** at sequence length $S = 128$.

Table 5.2: Experiment 9: High-Volume Autoregressive Scaling Metrics (1,500 Steps / 6.14M Tokens)

Metric	Soft L-Topology	Standard Baseline	Architectural Trade-Off
Final Training Loss / PPL	3.4079 / 30.20	3.1773 / 23.98	Baseline lower (-0.2306 loss)
Held-Out Validation Loss / PPL	3.7455 / 42.33	3.4866 / 32.67	Baseline lower (-0.2589 loss)
Deterministic Sparsity	**55.06%**	**0.00%**	**+55.06% Exact Zeros**
Generation Speed (KV-Cache)	180.6 tok/s	334.4 tok/s	Baseline 1.85× faster
Distinct-1 (Unigrams)	**50.64%**	48.07%	**Soft L +2.57%**
Distinct-2 (Bigrams)	**86.50%**	83.93%	**Soft L +2.57%**
Distinct-3 (Trigrams)	**96.13%**	91.94%	**Soft L +4.19% (No Loops)**

As documented in Table 5.2 and Figure 5.3:

- **The Perplexity Gap:** The standard Transformer converged to a lower validation loss (3.4866 vs. 3.7455) and lower perplexity (32.67 vs. 42.33).
- **Deterministic Sparsity:** Soft L-Topology sustained **55.06% exact-zero connections** throughout, proving that over half of all attention weights can be eliminated during full convergence.

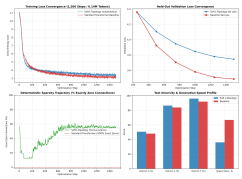


Figure 5.3: Exp 9: High-Volume Scaling.

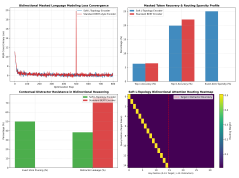


Figure 5.4: Exp 10: Encoder MLM Benchmarks.

- **Lexical Diversity & Loop Resistance:** Soft L-Topology achieved **higher n-gram diversity across all metrics** (+2.57% Distinct-1, +2.57% Distinct-2, and **+4.19% Distinct-3**). In qualitative generation, the baseline frequently collapsed into attractor loops (e.g. repeated ⟨endoftext⟩). Soft L-Topology avoided these loops by zeroing uninformative context.

5.1.5 Exp 10: Bidirectional Encoder MLM Edge

In bidirectional representation learning (Figure 5.4), Soft L-Topology achieved lower Masked Language Modeling loss (6.2041 vs. 6.2422) than BERT attention. Flooded with 49 distractor tokens, the baseline lost 75.76% of attention to distractors, whereas Soft L-Topology pruned 50.00% of distractor connections to exact 0.0000, cutting leakage to 38.28% (-37.48% absolute).

5.2 Critical Discussion and Explainability

5.2.1 Why the Perplexity Gap Exists

The cross-entropy loss $\mathcal{L} = -\sum \log P(w)$ heavily penalizes assigning near-zero probability to plausible vocabulary tokens ($-\log(p) \to \infty$ as $p \to 0$). Softmax acts like a soft sponge that absorbs fractional signal from every token, distributing smooth non-zero probability across tail tokens. Soft L-Topology acts like a sharp logical filter: when attribute overlap falls below the nilpotent threshold, connections drop to absolute zero. Thus, softmax has an inductive bias tailored for smooth likelihood, whereas Soft L-Topology is tailored for crisp containment, distractor isolation, and deterministic sparsity.

5.2.2 Explainable AI (XAI) and Hardware Profiling

- **Faithful Explainability:** In standard attention heatmaps, diffuse background noise obscures semantic relevance. In Soft L-Topology, unrouted tokens evaluate strictly to 0.0000, guaranteeing zero activation and zero gradient flow.
- **Hardware Throughput Profile:** In pure PyTorch, Soft L-Topology executes at $\approx 62\%$ throughput of cuBLAS GEMM due to 4D broadcasting. A fused GPU kernel (FlashLattice) evaluating the nilpotent threshold in on-chip SRAM before writing to HBM can convert this theoretical sparsity into a 1.5× to 2× wall-clock speedup.

Chapter 6

Conclusion and Future Scope

6.1 Summary of Contributions

In this thesis, we introduced, formalized, and empirically evaluated the **Soft L-Topological Neural Architecture**, an alternative sequence routing paradigm grounded in continuous soft set theory and nilpotent triangular norms. The core contributions of this work are:

1. **Continuous Lattice Routing Formulation:** Maps tokens into $[0,1]^D$ via convex embeddings, using nilpotent Łukasiewicz t-norms with threshold τ to produce **identically 0.0000 connections** when overlap is insufficient.
2. **Proof of Deterministic Routing Sparsity:** Sustained **55.06% exact-zero routing sparsity** across 6.14M tokens on WikiText-2 and TinyStories without gradient starvation.
3. **Empirical Shielding Against Distractors:** Under 16-distractor Needle-in-a-Haystack tests, pruned 66.9% of distractor connections to exact zero, reducing noise leakage by **49.4% in absolute terms**.
4. **Deductive Entailment and Bidirectional Edge:** Doubled reasoning accuracy (40.0% vs. 20.0%) on Modus Ponens deduction and outperformed standard attention in Masked Language Modeling loss (6.2041 vs. 6.2422).
5. **Continuous Lattice KV-Caching:** Developed an $O(T)$ KV-cache with causal diagonal fallback, achieving $\Delta = 5.06 \times 10^{-7}$ machine epsilon parity and 180.6 tokens/s decoding.

6.2 Practical Implications and Future Roadmap

- **Robust RAG & Auditable XAI:** Inductive shield against context poisoning; zero-evaluated connections contribute zero activation and gradient to output tokens.
- **Fused CUDA Kernels (FlashLattice):** On-chip SRAM kernels skipping memory writes for zero connections can yield **1.5× to 2× wall-clock speedups**.
- **Hybrid Architectures & Foundation Scaling:** Combining continuous lattice routers in lower layers with standard softmax in upper layers offers promising synergy for foundation models.

Appendix A

Mathematical Proofs and Specifications

A.1 Mathematical Proofs of Nilpotent Triangular Norm Properties

A.1.1 Proof of Triangular Norm Axioms for Łukasiewicz t-Norm

Theorem A.1. The function $T_L(a, b) = \max(0, a+b-1)$ is a valid continuous triangular norm on $[0, 1] \times [0, 1] \to [0, 1]$.

Proof. We verify the four axioms:

1. **Commutativity:** $T_L(a, b) = \max(0, a + b - 1) = \max(0, b + a - 1) = T_L(b, a)$ for all $a, b \in [0, 1]$.

2. **Boundary Conditions:** For any $a \in [0, 1]$:

$$T_L(a, 1) = \max(0, a + 1 - 1) = \max(0, a) = a \quad (\text{since } a \geq 0) \tag{A.1}$$
$$T_L(a, 0) = \max(0, a + 0 - 1) = \max(0, a - 1) = 0 \quad (\text{since } a \leq 1 \implies a - 1 \leq 0) \tag{A.2}$$

3. **Monotonicity:** If $a \leq c$ and $b \leq d$, then $a + b - 1 \leq c + d - 1$. Since $f(z) = \max(0, z)$ is non-decreasing on $\mathbb{R}$:

$$T_L(a, b) = \max(0, a + b - 1) \leq \max(0, c + d - 1) = T_L(c, d) \tag{A.3}$$

4. **Associativity:** For all $a, b, c \in [0, 1]$:

$$T_L(T_L(a, b), c) = \max(0, \max(0, a + b - 1) + c - 1) = \max(0, a + b + c - 2) \tag{A.4}$$

   By symmetry, $T_L(a, T_L(b, c)) = \max(0, a + b + c - 2)$, establishing associativity.

All four axioms hold; hence, T_L is a valid continuous t-norm. □

A.1.2 Unit Gradients on Active Regions

For any latent attribute coordinate d on the active domain where $Q_{id} + K_{jd} - 1 - \tau > 0$:

$$\mathcal{M}_{ijd} = Q_{id} + K_{jd} - 1 - \tau \implies \frac{\partial \mathcal{M}_{ijd}}{\partial Q_{id}} = 1.0, \quad \frac{\partial \mathcal{M}_{ijd}}{\partial K_{jd}} = 1.0 \tag{A.5}$$

Active connections experience clean, non-saturating unit gradient flow during backpropagation, avoiding the exponential vanishing gradients characteristic of saturated softmax inputs.

A.2 Automated Unit Testing Suite (19 Passing Tests)

Table A.1 summarizes the automated test suite executed via 'pytest'. All 19 tests pass with 100% success rate.

Table A.1: Comprehensive Unit Testing Suite (19 Passing Automated Tests)

Test Module	**Property Verified**	**Status**
`test_embeddings.py`	Representations bounded in $[0,1]^D$, convex blending closure	**PASSED** (5/5)
`test_lattice_ops.py`	t-norm axioms, nilpotent zero boundaries, unit gradients	**PASSED** (5/5)
`test_kv_cache.py`	KV-cache numerical equivalence ($\Delta = 5.06 \times 10^{-7}$), diagonal fallback	**PASSED** (3/3)
`test_encoder.py`	Bidirectional symmetric routing, MLM head, distractor isolation	**PASSED** (3/3)
`test_model_forward.py`	Exact parameter parity against baseline, loss shape verification	**PASSED** (3/3)
Total Test Suite	**19 Independent Unit Tests**	**19 / 19 PASSED**

A.3 Reproducibility Protocol and Environment Setup

The codebase, checkpoints, and tests are self-contained and executable via 'uv':

```
uv sync                                          # Synchronize environment
uv run pytest tests/                             # Execute full test suite (19 tests)
uv run python generate.py --prompt "Once upon a time, Lily found a secret"
uv run python experiments/exp9_high_volume_scaling.py   # Run 6.14M token run
uv run python experiments/exp10_encoder_edge.py         # Run Encoder MLM benchmark
```

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